higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

T530**(E)**(M28)T

NATIONAL CERTIFICATE

ELECTROTECHNICS N4

(8080074)

28 March 2019 (X-Paper)
09:00–12:00

Calculators may be used.

This question paper consists of 6 pages and one formula sheet of 2 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
ELECTROTECHNICS N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Questions may be answered in any order but subsections of questions must be kept together.
5. Write neatly and legibly.

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QUESTION 1

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–K) next to the question number (1.1–1.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.1	Atom which has lost or gained one or more electrons	A	magnetic flux
1.2	Algebraic sum of voltage drops in any closed circuit is equal to the algebraic sum of the applied emf	B	Lenz's law
1.3	Magnetic field or entire quantity of magnetic lines surrounding a magnet, taken as whole	C	electrolytic type
1.4	Opposition offered to the establishment of magnetic lines of force in a magnet circuit	D	magnetising force
1.5	Emf set up in the core of rotating electrical machinery or core of transformers, contactors, et cetera causes currents to circulate in them	E	ion
1.6	Direction of an induced emf tends to set up a current opposing the motion or the change of flux responsible for that emf	F	reluctance
1.7	When the change in flux conditions is due to changing current conditions within the conductor itself	G	self-induction
1.8	Somewhat similar in construction to paper-dielectric capacitors, but the paper used is an absorbent type impregnated with a solution containing borax	H	back emf
1.9	Voltage that opposes the change in current which induced it	I	ampere
1.10	$H = \frac{NI}{l} A/m$	J	Kirchhoff's second law
		K	eddy currents

(10 × 1) **[10]**

QUESTION 2

2.1 Why are certain currents that circulate in transformer cores undesirable and what are they called? (4)

2.2 What are the properties of hard and soft magnetic materials and where are they used? (4)

2.3 Direct current (DC) machines are named according to the method used to connect the field coils.

Name TWO main types of DC machines. (2)

2.4 What is meant by a *positive temperature coefficient of resistance*? (2)

2.5 A resistor of 180 ohms is connected in parallel with a resistance of 20 ohms. This combination is then connected in series with an unknown resistance. The circuit is then connected across a 60 V DC supply.

Calculate the following:

2.5.1 Value of the unknown resistor when 2 A current is drawn from the supply (4)

2.5.2 Total resistance of the circuit (1)

2.5.3 Power dissipated in the circuit (1)

[18]

QUESTION 3

3.1 Name THREE principal EMF sources. (3)

3.2 What is meant by *EMF of a battery*? (2)

3.3 In a magnet each line of force passes from the south pole to the north pole.

What are the FIVE characteristics of these lines of force? (5)

3.4 Two batteries are connected in parallel and must be charged from a DC generator having an open-circuit voltage of 28 V and an internal resistance of 0,2 ohms. The open-circuit voltages of the batteries are 20 V and 16 V with an internal resistance of 0,5 ohms and 0,7 ohms respectively.

Use Kirchhoff's laws to determine the initial charging currents of the batteries. (8)

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QUESTION 4

4.1 What are the FOUR methods of improving commutation in a DC machine? (4)

4.2 Name the TWO principal laws of electromagnetism and state each law. (3)

4.3 What effect will an increase in temperature have on the resistance of the following materials? Also indicate whether it will have a negative or positive temperature coefficient.

4.3.1 Copper

4.3.2 Carbon

(2 × 2) (4)

4.4 How can a magnetic field of a solenoid be strengthened? (2)

4.5 Calculate the resistance of a starting resistor required to start a 440 V DC shunt motor with an input of 10 KW. The maximum armature current is not to exceed 1,5 times the full-load armature current. The armature has a resistance of 0,67 ohms and the field resistance is 161 ohms. (5)

[18]

QUESTION 5

5.1 Name FOUR common methods of cooling transformers. (4)

5.2 At what speed must a SIX-pole alternator be driven to produce an EMF with a frequency of 60 Hz? (2)

5.3 How is a compound DC generator described by means of a load characteristic? (6)

5.4 A moving-coil instrument has a resistance of 75 ohms and requires 300 mA to give a full-scale deflection.

Calculate the value of the additional component necessary to enable the instrument to measure a voltage of 60 V. (6)

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QUESTION 6

6.1 What is the function of Buchholz's device on a transformer? (2)

6.2 Name TWO purposes of the capacitor on a capacitor-start single-phase induction motor. (2)

6.3 What is the purpose of crystals commonly used in the breather of a transformer? (2)

6.4 Name TWO types of feeders used in transmission systems. (2)

6.5 In which system does failure of ONE interconnecting feeder interrupt the supply to any of the other substations? (1)

6.6 What are the TWO main types of connection in a three-phase induction motor? (2)

6.7 State any FOUR advantages of an induction motor compared to DC motor. (4)

6.8 Give THREE disadvantages of coal-fired power stations. (3)

[18]

TOTAL: 100

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ELECTROTECHNICS N4

FORMULA SHEET

Any applicable formula may also be used.

1. PRINCIPLES OF ELECTRICITY

$$E = V + Ir$$

$$V = IR$$

$$R_{se} = R_1 + R_2 + ... R_n$$

$$R_p = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + ... \frac{1}{R_n}}$$

$$R = \rho \frac{\ell}{a}$$

$$\frac{R_1}{R_2} = \frac{1 + \alpha_o T_1}{1 + \alpha_o T_2}$$

$$R_t = R_\theta \, [1 + \alpha_\theta \, (t - \theta)]$$

$$P = VI = I^2 R = \frac{V^2}{R}$$

$$\Phi = \frac{mmf}{S} = \frac{IN}{S}$$

$$H = \frac{IN}{\ell}$$

$$F = B\ell I$$

$$E = \frac{\Delta\Phi}{\Delta t} . N$$

$$E = B\ell v$$

$$E = \frac{L\Delta I}{\Delta t}$$

$$L = \frac{\Delta\Phi}{\Delta I} . N$$

$$Q = VC$$

$$Q_{se} = Q_t = Q_1 = Q_2 \, ... = Q_n$$

$$C_{se} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + ... \frac{1}{C_n}}$$

$$Q_p = Q_1 + Q_2 + ... Q_n$$

$$C_p = C_1 + C_2 + ... C_n$$

2. DIRECT-CURRENT MACHINES

$$E = \frac{2Z}{c} . \frac{Np}{60} . \Phi$$

$$c = 2a$$

$$E_{gen} = V + I_a \, R_a$$

$$E_{mot} = V - I_a \, R_a$$

$$R_{start} = \frac{(V - E)}{I_a} - R_a$$

3. ALTERNATING-CURRENT MACHINES

$$E_m = 2\pi BANn$$

$$e = E_m \sin (2\pi f . \, t \times 57{,}3)°$$

$$E_{ave} = 0{,}637 \, E_m$$

$$E_{rms} = 0{,}707 \, E_m$$

$$T = \frac{1}{f}$$

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$$f = \frac{Np}{60}$$

$$\omega = 2\pi f$$

$$Z_L = R + j\omega L$$

$$Z_c = R - j\frac{1}{\omega C}$$

$$pf = \cos\phi = \frac{R}{Z}$$

$$S = VI$$

$$P = V.I\cos\phi = I^2R$$

$$Q = V.I\sin\phi$$

4. TRANSFORMERS

$$E = 4{,}44\ f\ \Phi_m\ N$$

$$k_t = \frac{N_1}{N_2} = \frac{V_1}{V_2} = \frac{I_2}{I_1}$$

5. MEASURING INSTRUMENTS

$$R_{SH} = \frac{i_m\ R_m}{I_{sh}}$$

$$R_{se} = \frac{V}{i_m} - R_m$$

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